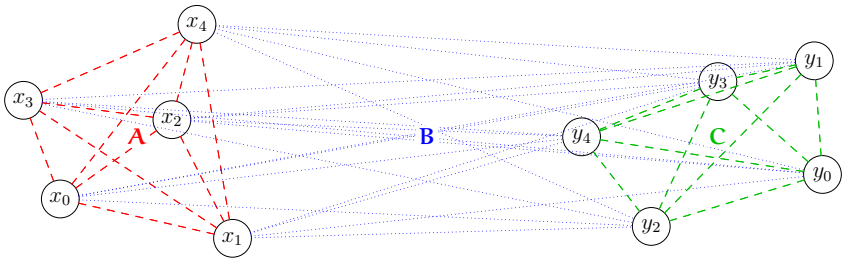
Mathematics of Machine Learning

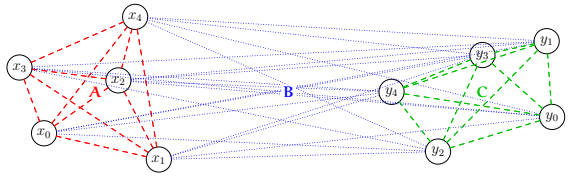
Gaussian Integrals



Gaussian integrals, completing the square, moment generating functions, marginals, conditionals, Kalman filter

Outline

1. **One Dimension**
2. Many Dimensions
3. Marginals and Conditionals
4. Inference with Gaussians



Why Gaussians?

- Gaussians (normal distributions) are everywhere in machine learning: Gaussian processes, Kalman filters, linear regression, the noise in diffusion models, . . .
- The reason is that we can do all their integrals exactly
- Marginalising, conditioning, adding or linearly transforming Gaussians gives another Gaussian
- The algebra is long-winded, but it uses only three tricks: the basic Gaussian integral, completing the square, and diagonalising a matrix
- Know the results and when to use them; the derivations are there so you can trust them

The Gaussian Integral

- We cannot find an antiderivative of $\mathrm{e}^{-x^2/2}$, but there is a trick: square the integral

$$I^2 = \int_{-\infty}^{\infty} \mathrm{e}^{-x^2/2} \mathrm{d}x \int_{-\infty}^{\infty} \mathrm{e}^{-y^2/2} \mathrm{d}y = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mathrm{e}^{-(x^2+y^2)/2} \mathrm{d}x\mathrm{d}y$$

- In polar coordinates ($x^2 + y^2 = r^2$, $\mathrm{d}x\mathrm{d}y = r\mathrm{d}r\mathrm{d}\theta$)

$$I^2 = \int_0^{2\pi} \mathrm{d}\theta \int_0^{\infty} r\mathrm{e}^{-r^2/2} \mathrm{d}r = 2\pi \int_0^{\infty} \mathrm{e}^{-u} \mathrm{d}u = 2\pi$$

  using $u = r^2/2$, $\mathrm{d}u = r\mathrm{d}r$

- So

$$\int_{-\infty}^{\infty} \mathrm{e}^{-x^2/2} \mathrm{d}x = \sqrt{2\pi}$$

Normal Distribution

- Changing variables, $z = (x - \mu)/\sigma$ so $\mathrm{d}x = \sigma \mathrm{d}z$,

$$\int_{-\infty}^{\infty} \mathrm{e}^{-(x-\mu)^2/(2\sigma^2)} \mathrm{d}x = \sigma \int_{-\infty}^{\infty} \mathrm{e}^{-z^2/2} \mathrm{d}z = \sqrt{2\pi}\sigma$$

- So the normal density

$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \mathrm{e}^{-(x-\mu)^2/(2\sigma^2)}$$

  integrates to one

- (For small σ the density exceeds one at $x = \mu$: densities are not probabilities)

Completing the Square

- Any quadratic in the exponent can be rewritten (for $a > 0$)

$$-\frac{a}{2}x^2 + bx = -\frac{a}{2}\left(x - \frac{b}{a}\right)^2 + \frac{b^2}{2a}$$

- So the integral is a shifted Gaussian times a constant

$$\int_{-\infty}^{\infty} \mathrm{e}^{-\frac{a}{2}x^2 + bx}\,\mathrm{d}x = \mathrm{e}^{\frac{b^2}{2a}} \int_{-\infty}^{\infty} \mathrm{e}^{-\frac{a}{2}\left(x - \frac{b}{a}\right)^2}\,\mathrm{d}x = \sqrt{\frac{2\pi}{a}}\,\mathrm{e}^{\frac{b^2}{2a}}$$

- This one result does most of the work in this lecture

Moment Generating Function

- The **moment generating function** of X is $M(t) = \mathbb{E}\left[\mathrm{e}^{tX}\right]$
- For $X \sim \mathcal{N}(\mu,\sigma^2)$, completing the square gives

$$M(t) = \int_{-\infty}^{\infty} \mathrm{e}^{tx}\mathcal{N}(x \mid \mu,\sigma^2)\mathrm{d}x = \mathrm{e}^{\mu t+\frac{1}{2}\sigma^2 t^2}$$

- Expanding $\mathrm{e}^{tX} = 1 + tX + \frac{1}{2}t^2X^2 + \cdots$ shows $M^{(n)}(0) = \mathbb{E}[X^n]$

$$M'(0) = \mu, \qquad M''(0) = \mu^2 + \sigma^2$$

- The cumulant generating function $\log M(t) = \mu t + \frac{1}{2}\sigma^2 t^2$ stops at t^2: a Gaussian has no cumulants beyond the mean and variance

Sums of Gaussians

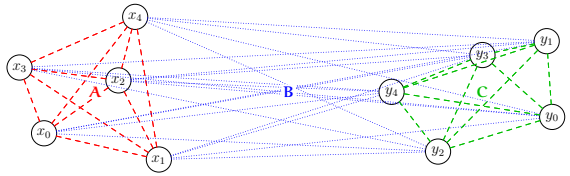
- A moment generating function determines its distribution
- If X and Y are independent, $\mathbb{E}\left[\mathrm{e}^{t(X+Y)}\right] = \mathbb{E}\left[\mathrm{e}^{tX}\right]\mathbb{E}\left[\mathrm{e}^{tY}\right]$
- So for $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$

$$\mathbb{E}\left[\mathrm{e}^{t(X+Y)}\right] = \mathrm{e}^{(\mu_1+\mu_2)t+\frac{1}{2}(\sigma_1^2+\sigma_2^2)t^2}$$

  which is the MGF of $\mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$
- Means add and variances add, and the sum is still Gaussian
- No integral needed: this is why MGFs are so useful

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Independent Variables

- With $\boldsymbol{x} = (x_1,\ldots,x_n)^\mathsf{T}$ and $\|\boldsymbol{x}\|^2 = \sum_i x_i^2$, the exponential of a sum is a product of exponentials

$$\int \mathrm{e}^{-\frac{1}{2}\|\boldsymbol{x}\|^2}\mathrm{d}\boldsymbol{x} = \prod_{i=1}^{n}\int_{-\infty}^{\infty}\mathrm{e}^{-x_i^2/2}\mathrm{d}x_i = (2\pi)^{n/2}$$

- Similarly with a different variance λ_i in each direction

$$\int \mathrm{e}^{-\frac{1}{2}\sum_i x_i^2/\lambda_i}\mathrm{d}\boldsymbol{x} = \prod_{i=1}^{n}\sqrt{2\pi\lambda_i}$$

- The general case can be rotated into this one

The Multivariate Normal

- For a positive definite covariance $\boldsymbol{\Sigma} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^\mathsf{T}$ (eigenvalues $\lambda_i > 0$) consider $I = \int \mathrm{e}^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\mathsf{T}\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})}\mathrm{d}\boldsymbol{x}$

- Rotate, $\boldsymbol{y} = \boldsymbol{V}^\mathsf{T}(\boldsymbol{x}-\boldsymbol{\mu})$: the Jacobian $\boldsymbol{V}$ has $|\det \boldsymbol{V}| = 1$ (rotations keep volumes)

$$I = \int \mathrm{e}^{-\frac{1}{2}\boldsymbol{y}^\mathsf{T}\boldsymbol{\Lambda}^{-1}\boldsymbol{y}}\mathrm{d}\boldsymbol{y} = \prod_i \sqrt{2\pi\lambda_i} = \sqrt{\det(2\pi\boldsymbol{\Sigma})}$$

  as $\det\boldsymbol{\Sigma} = \prod_i \lambda_i$ and $\det(c\boldsymbol{\Sigma}) = c^n \det\boldsymbol{\Sigma}$

- So the multivariate normal density is

$$\mathcal{N}(\boldsymbol{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{\det(2\pi\boldsymbol{\Sigma})}} \mathrm{e}^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\mathsf{T}\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})}$$

Completing the Square in Many Dimensions

- For symmetric positive definite $\boldsymbol{A}$

$$-\frac{1}{2}\boldsymbol{x}^\mathsf{T}\boldsymbol{A}\boldsymbol{x}+\boldsymbol{b}^\mathsf{T}\boldsymbol{x}=-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{A}^{-1}\boldsymbol{b})^\mathsf{T}\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{A}^{-1}\boldsymbol{b})+\frac{1}{2}\boldsymbol{b}^\mathsf{T}\boldsymbol{A}^{-1}\boldsymbol{b}$$

  (multiply out the right-hand side to check)

- So, by the last slide with $\boldsymbol{\Sigma}=\boldsymbol{A}^{-1}$,

$$\int \mathrm{e}^{-\frac{1}{2}\boldsymbol{x}^\mathsf{T}\boldsymbol{A}\boldsymbol{x}+\boldsymbol{b}^\mathsf{T}\boldsymbol{x}}\,\mathrm{d}\boldsymbol{x}=\sqrt{\frac{(2\pi)^n}{\det\boldsymbol{A}}}\,\mathrm{e}^{\frac{1}{2}\boldsymbol{b}^\mathsf{T}\boldsymbol{A}^{-1}\boldsymbol{b}}$$

- This is the central result: every Gaussian integral in this course is an instance of it

Multivariate Moment Generating Function

- For $\boldsymbol{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the central result with $\boldsymbol{A} = \boldsymbol{\Sigma}^{-1}$ gives

$$M(\boldsymbol{t}) = \mathbb{E}\left[\mathrm{e}^{\boldsymbol{t}^{\mathsf{T}}\boldsymbol{x}}\right] = \mathrm{e}^{\boldsymbol{t}^{\mathsf{T}}\boldsymbol{\mu} + \frac{1}{2}\boldsymbol{t}^{\mathsf{T}}\boldsymbol{\Sigma}\boldsymbol{t}}$$

- Differentiating at $\boldsymbol{t} = \boldsymbol{0}$

$$\mathbb{E}[\boldsymbol{x}] = \boldsymbol{\mu}, \qquad \mathbb{E}\left[(\boldsymbol{x} - \boldsymbol{\mu})(\boldsymbol{x} - \boldsymbol{\mu})^{\mathsf{T}}\right] = \boldsymbol{\Sigma}$$

- A Gaussian is completely described by its mean and its covariance

Linear Transformations

- Let $\boldsymbol{z} = \boldsymbol{W}\boldsymbol{x} + \boldsymbol{c}$ with $\boldsymbol{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$
- Its MGF is

$$\mathbb{E}\left[\mathrm{e}^{\boldsymbol{t}^\mathsf{T}\boldsymbol{z}}\right] = \mathrm{e}^{\boldsymbol{t}^\mathsf{T}\boldsymbol{c}}\mathbb{E}\left[\mathrm{e}^{(\boldsymbol{W}^\mathsf{T}\boldsymbol{t})^\mathsf{T}\boldsymbol{x}}\right] = \mathrm{e}^{\boldsymbol{t}^\mathsf{T}(\boldsymbol{W}\boldsymbol{\mu}+\boldsymbol{c})+\frac{1}{2}\boldsymbol{t}^\mathsf{T}\boldsymbol{W}\boldsymbol{\Sigma}\boldsymbol{W}^\mathsf{T}\boldsymbol{t}}$$

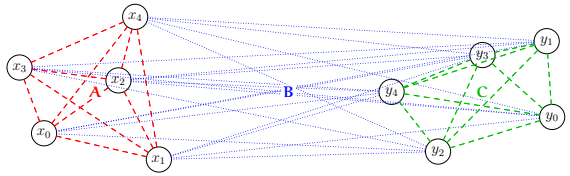
- So

$$\boldsymbol{z} \sim \mathcal{N}\left(\boldsymbol{W}\boldsymbol{\mu} + \boldsymbol{c},\, \boldsymbol{W}\boldsymbol{\Sigma}\boldsymbol{W}^\mathsf{T}\right)$$

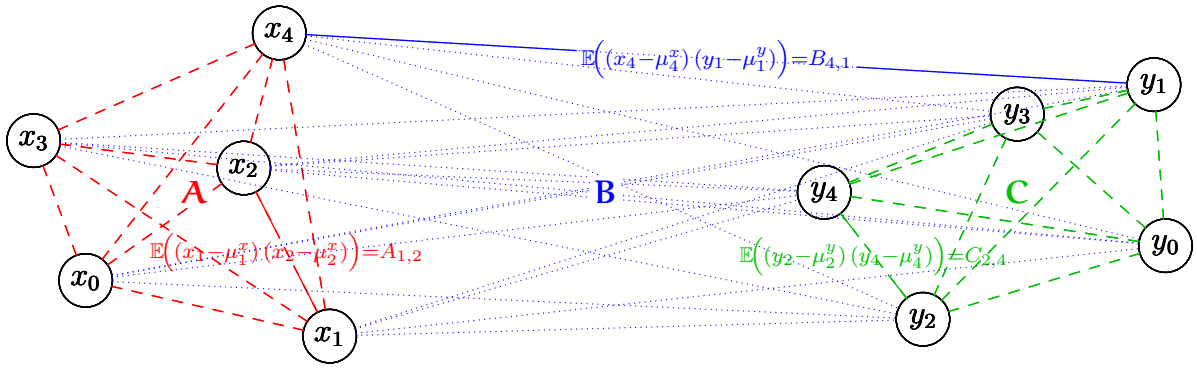
- Linear maps of Gaussians are Gaussian, and we only need to track the mean and covariance ($\boldsymbol{W}$ need not be square)

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Joint Gaussians



$$\begin{pmatrix} \boldsymbol{z}_1 \\ \boldsymbol{z}_2 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}, \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix} \right), \qquad \boldsymbol{\Sigma}_{21} = \boldsymbol{\Sigma}_{12}^{\mathsf{T}}$$

Marginalising

- We want $p(\boldsymbol{z}_1) = \int p(\boldsymbol{z}_1, \boldsymbol{z}_2) \mathrm{d}\boldsymbol{z}_2$
- Doing the integral directly takes a page of block-matrix algebra (see the notes)
- Instead take $\boldsymbol{t} = (\boldsymbol{t}_1, \boldsymbol{0})$ in the MGF, so that $\boldsymbol{t}^\mathsf{T}\boldsymbol{z} = \boldsymbol{t}_1^\mathsf{T}\boldsymbol{z}_1$

$$\mathbb{E}\left[\mathrm{e}^{\boldsymbol{t}_1^\mathsf{T}\boldsymbol{z}_1}\right] = \mathrm{e}^{\boldsymbol{t}_1^\mathsf{T}\boldsymbol{\mu}_1 + \frac{1}{2}\boldsymbol{t}_1^\mathsf{T}\boldsymbol{\Sigma}_{11}\boldsymbol{t}_1}$$

- That is the MGF of $\mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$, so

$$\boldsymbol{z}_1 \sim \mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$$

- To marginalise, just strike out the rows and columns of the variables you integrate away

Conditioning: a Light Derivation

- We want $p(\boldsymbol{z}_1 \mid \boldsymbol{z}_2)$
- Subtract the best linear prediction of $\boldsymbol{z}_1$ from $\boldsymbol{z}_2$

$$\boldsymbol{w} = \boldsymbol{z}_1 - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{z}_2$$

- $\boldsymbol{w}$ is a linear map of $\boldsymbol{z}$, so $(\boldsymbol{w}, \boldsymbol{z}_2)$ are jointly Gaussian, and

$$\mathrm{Cov}(\boldsymbol{w}, \boldsymbol{z}_2) = \boldsymbol{\Sigma}_{12} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{22} = 0$$

- For jointly Gaussian variables, uncorrelated means independent (the MGF factorises)
- So knowing $\boldsymbol{z}_2$ tells us nothing about $\boldsymbol{w}$, and $\boldsymbol{z}_1 = \boldsymbol{w} + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{z}_2$ with $\boldsymbol{z}_2$ now fixed

The Conditional Distribution

- From the linear-transformation rule, $\boldsymbol{w}$ has mean $\boldsymbol{\mu}_1 - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\mu}_2$ and covariance $\boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$
- Hence

$$\boldsymbol{z}_1 \mid \boldsymbol{z}_2 \sim \mathcal{N}\big(\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\boldsymbol{z}_2 - \boldsymbol{\mu}_2),\, \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}\big)$$

- The mean moves linearly with what we observe
- The covariance shrinks, by an amount that does not depend on the value we observe
- This one formula is the engine behind Gaussian processes and Kalman filters

The Long Way Round

- The same results follow by brute force from the block factorisation

$$\begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix} = \begin{pmatrix} \mathbf{I} & \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \boldsymbol{S} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Sigma}_{22} \end{pmatrix} \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21} & \mathbf{I} \end{pmatrix}.$$

  where $\boldsymbol{S} = \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$ (the **Schur complement**).

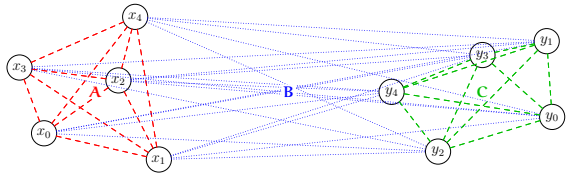
- Inverting each factor splits the exponent into a part for $\boldsymbol{z}_2$ and a part for $\boldsymbol{z}_1$ given $\boldsymbol{z}_2$.

- It also gives $\det\boldsymbol{\Sigma} = \det\boldsymbol{S}\det\boldsymbol{\Sigma}_{22}$.

- The full derivation is in the notes: worth reading once, not memorising.

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A Noisy Measurement

- Prior belief about a quantity: $x \sim \mathcal{N}(\mu_0, s_0^2)$
- We measure $y = x + \epsilon$ with noise $\epsilon \sim \mathcal{N}(0, \sigma^2)$
- (x, y) is a linear map of (x, ϵ), so jointly Gaussian with

$$\begin{pmatrix} x \\ y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_0 \\ \mu_0 \end{pmatrix}, \begin{pmatrix} s_0^2 & s_0^2 \\ s_0^2 & s_0^2 + \sigma^2 \end{pmatrix} \right)$$

- Conditioning on y

$$\mathbb{E}[x \mid y] = \mu_0 + \frac{s_0^2}{s_0^2 + \sigma^2}(y - \mu_0), \qquad \frac{1}{\mathbb{V}\mathrm{ar}(x \mid y)} = \frac{1}{s_0^2} + \frac{1}{\sigma^2}$$

- The estimate moves towards the measurement, and the precisions (inverse variances) add

Combining Sensors

- Two sensors measure the same x, with noise variances σ_1^2 and σ_2^2, giving y_1 and y_2
- With a flat prior ($s_0 \to \infty$), applying the last result twice gives

$$\mathbb{E}[x \mid y_1, y_2] = \frac{y_1/\sigma_1^2 + y_2/\sigma_2^2}{1/\sigma_1^2 + 1/\sigma_2^2}, \qquad \frac{1}{\mathbb{V}\mathrm{ar}(x \mid y_1, y_2)} = \frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}$$

- A precision-weighted average: trust the better sensor more
- With n equally good measurements the variance falls as σ^2/n

Predicting One Variable from Another

- Two correlated quantities (e.g. height and weight) with correlation ρ

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix} \right)$$

- Conditioning on x_1

$$\mathbb{E}[x_2 \mid x_1] = \mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x_1 - \mu_1), \qquad \mathbb{V}\mathrm{ar}(x_2 \mid x_1) = \sigma_2^2(1-\rho^2)$$

- The prediction is a straight line: linear regression

- As $|\rho| < 1$ the prediction is fewer standard deviations from the mean than x_1 is: **regression to the mean**

Tracking: the Kalman Filter

- A position drifts, $x_{t+1} = x_t + \eta_t$ with $\eta_t \sim \mathcal{N}(0,q)$, and is observed as $y_t = x_t + \epsilon_t$ with $\epsilon_t \sim \mathcal{N}(0,\sigma^2)$
- Our belief stays Gaussian, $x_t \sim \mathcal{N}(m_t,v_t)$. **Predict** (a sum of Gaussians): $x_{t+1} \sim \mathcal{N}(m_t,\, v_t + q)$
- **Update:** a noisy measurement, with prior $\mathcal{N}(m_t,v_t + q)$

$$m_{t+1} = m_t + k(y_{t+1} - m_t), \qquad v_{t+1} = (1-k)(v_t + q),$$

$$k = \frac{v_t + q}{v_t + q + \sigma^2}$$

- That is the Kalman filter (with vectors and matrices it guides aircraft and phones)

Towards Gaussian Processes

- Suppose the function values at some inputs are jointly Gaussian, and we observe some of them with noise

- Put the unknown value f_* in $\boldsymbol{z}_1$ and the observations $\boldsymbol{y}$ in $\boldsymbol{z}_2$

$$\begin{pmatrix} f_* \\ \boldsymbol{y} \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{pmatrix} k_* & \boldsymbol{k}_*^\mathsf{T} \\ \boldsymbol{k}_* & \mathbf{K} + \sigma^2\mathbf{I} \end{pmatrix}\right)$$

- The conditional formula gives the prediction immediately

$$f_* \mid \boldsymbol{y} \sim \mathcal{N}\big(\boldsymbol{k}_*^\mathsf{T}(\mathbf{K} + \sigma^2\mathbf{I})^{-1}\boldsymbol{y},\, k_* - \boldsymbol{k}_*^\mathsf{T}(\mathbf{K} + \sigma^2\mathbf{I})^{-1}\boldsymbol{k}_*\big)$$

- Where the covariances come from is the subject of the next lecture

Summary

- The toolkit
  - ⋆ $\int \mathrm{e}^{-\frac{1}{2}\boldsymbol{x}^\mathsf{T}\mathbf{A}\boldsymbol{x}+\boldsymbol{b}^\mathsf{T}\boldsymbol{x}}\mathrm{d}\boldsymbol{x} = \sqrt{(2\pi)^n/\det\mathbf{A}}\,\mathrm{e}^{\frac{1}{2}\boldsymbol{b}^\mathsf{T}\mathbf{A}^{-1}\boldsymbol{b}}$
  - ⋆ MGF: $\mathbb{E}\left[\mathrm{e}^{\boldsymbol{t}^\mathsf{T}\boldsymbol{x}}\right] = \mathrm{e}^{\boldsymbol{t}^\mathsf{T}\boldsymbol{\mu}+\frac{1}{2}\boldsymbol{t}^\mathsf{T}\boldsymbol{\Sigma}\boldsymbol{t}}$
  - ⋆ Linear maps: $\boldsymbol{W}\boldsymbol{x}+\boldsymbol{c} \sim \mathcal{N}(\boldsymbol{W}\boldsymbol{\mu}+\boldsymbol{c}, \boldsymbol{W}\boldsymbol{\Sigma}\boldsymbol{W}^\mathsf{T})$
  - ⋆ Marginals: strike out rows and columns
  - ⋆ Conditionals: $\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\boldsymbol{z}_2 - \boldsymbol{\mu}_2)$ and $\boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$
- Gaussians stay Gaussian, so inference reduces to linear algebra on means and covariances